BodyDay（身体的一天）· 作品开发文档

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	H5
	BodyDay ·
	2026 · · 4 H5

1.1 问题

— —

—— 2024 》

1.2 目标

BodyDay

1.
2. → →
3. →
4. →
5.

1.3 目标用户与场景

-
-
-

1.4 功能需求（P0 / P1）

P0 + → → /
/ / ID / / / →
/ / /

P1

AI

1.5 竞品对比与取舍

/

		BodyDay
---	---	---
Human Atlas ashemag/human-atlas	MIT © 2026 ashemag BodyParts3D CC BY 4.0	three 33 MB ID 3.11
Anatomy thebuggeddev/anatomy	LICENSE	
Vanatome / Z-Anatomy	MIT CC BY-SA 4.0	Z-Anatomy
BioDigital		
OpenStax	CC BY	S01–S05 S19

BodyDay

2.1 总体结构与产品形态

SPA HashRouter **16** 3.9

- React Three Fiber Three.js r186 + R3F 9 + Drei 10 Z-Anatomy GLB
- 240–280px / minmax(0,1fr) / 300–360px
- +
/ /
- ≤1024px ≤900px
- **`#/explore`** / **`#/detail/heart`** / **`#/about`** BodyDay /

+

→ + · + /

/ / /

/ +

DOM 240×3px 148×44px 100%

inert loading → ready-to-enter → entering → exploring /body /organ/* /cell/* /detail/* /about

2.2 功能模块层次

`

```
                  →    →                    inert

      BodyDay     →       /       /

├─       resolveView  URL →      body / system / organ / cell / detail / about

├─       ExploreWorkbench

|  ├─                    +

│  ├─       FullBodyStage / CellStage       WebGL

│  └─       KnowledgePaper

│     ├─          +              /       /          →

│     └─

└─       AboutPage  canvas = 0

         /    /    /          /    /

     organBridge

     anatomyModel        / atlas ID     /

     FullBodyStage          /       /         /

         /    /    /           /
```

`

2.3 跨尺度模型与「尺度的表达」

- → → / /
- →

2.4 数据设计

2.5 视觉与动效设计

+ #F7F5F1 #1A1917

V2 / / ↔

CC BY-SA 4.0 /about

public/models/ANATOMY_LICENSE.txt

Hover Selection Route / Camera Scale
src/design/interaction-tokens.css --bd-ease-out /
prefers-reduced-motion

3.1 前端分层与模块接口

---	---	---
	src/app/	resolveView / ROUTES / PAGE_INDEX
	src/domain/	
	src/integrations/	organBridge → /
	src/services/	atlas ID
	src/design/	
	src/three/	
	src/components/	UI
	src/features/workbench/	+ +
	src/features/startup/	StartupCover
	src/features/about/	AboutPage

三维舞台协议（`FullBodyStage` / `CellStage`）

---	---	---
	onReady + onProgress	+
	selectedStructureId + onSelectStructure	ID ID
	onStructureInfo	
	focusNonce focusTargetId	
	resetNonce	
	isolationMode	/
	visibleLayers	
	explodeProgress	+
	onProbe	

领域数据契约（`src/domain/types.ts`）

`

BodyEntity { id, nameZh, nameEn?, kind, parentId?, atlasStructureId?, proceduralId?,

summary, labels[], facts[], health[], sources[], reviewStatus, scaleNote? }

KnowledgeRelation { id, fromId, toId, relationType, description, sourceIds[],

confidence, reviewedAt }

GuidedLesson { id, title, theme, goal, entryEntityId, steps[], sourceIds[] }

LessonStep { id, kind, title, body, targetEntityId?, requiredInteraction?,

question?, sourceIds? }

QuizQuestion { id, prompt, options[], answerKey, explanation, sourceIds[] }

SourceRecord { id, publisher, title, url, accessDate, license?, scope }

`

`

rotate | focus:<entityId> | select:<entityId> | drilldown:<entityId> | answer

`

3.2 路由解析与页面索引

src/app/resolveView.ts URL body / system / organ / cell / detail / about ROUTES
PAGE_INDEX src/app/routes.ts AppShell <ExploreWorkbench view={view}/>
<Routes> Route React
→ props

showControls = !isCell && !entryIsolate

3.3 核心关系数据

14 **5**

---	---	---
part_of		→ →
functional		→
circulates_through		→
disease_context		→ →
extends_to		→

6. →
7. →
8. →

3.4 器官桥接：`organBridge` 人工核对表

src/integrations/organBridge.ts certainty verified / needs-review

- **verified** heart → heart cerebrum / cerebellum / brainstem → brain lungs → lungs spleen → spleen
- **needs-review** `target: null` liver kidneys pancreas eyeball intestine skin

target: null spleen pancreas needs-review

3.5 通用结构卡

src/domain/structureCard.ts

structureCardPath · · structureCardPlacement X / GLB

3.6 关键技术：自建三维舞台与真实拆解

V2 @vixotic/vanatome-react API select / focus / isolate / reset / visibleLayers React Three Fiber FullBodyStage GLB

atlas position

- body-shell 233
- nervous-system
- cerebrum / heart / spleen

GLB extras anatomyId / anatomyParentId / anatomySystem regional-anatomy=0 muscular=1 skeletal=2 cardiovascular/lymphatic/respiratory/digestive/urinary=3 nervous/endocrine/reproductive=4 = × 0.17 body-shell 0⟶0.4 ≥0.4 0⟶1⟶0⟶1 **967** 984 − 17 ⟶ 42% **734**

0.07 [0,1] canvas wheel OrbitControls @media (hover: hover) and (pointer: fine) display:none

reproductive 14 2 2 17 984 ⟶ 967

ID **`body-shell`**

3.7 关键技术：程序化细胞几何（原创）

DOM ✦

Tab

3.8 加载与就绪链路、返回上下文

/ GLB onProgress onReady

loading → ready-to-enter → entering → exploring

`ExploreReturnContext` src/features/workbench/returnContext.ts

ReturnSnapshot / /

Esc

3.9 界面设计说明（16 页口径）

---	---	---	---	---
01	·	/		+ /
02		/body		/
03		/systems/nervous-system		
04		/systems/cardiovascular-system		
05		/systems/lymphatic-system		
06		/organ/brain		
07		/organ/heart		/ /
08		/organ/spleen		
09		/cell/neuron		/ / /
10		/cell/leukocyte		/ /
11	·	/about	—	canvas = 0
12		/explore		
13	·	/detail/heart		
14	·	/detail/brain		
15	·	/detail/lungs		/
16	·	/detail/spleen		

/about 15 /about canvas = 0 16 10 20

3.10 内容资产：跨尺度科普脚本与检验题（保留，不作模式）

GuidePanel QuizBlock 3 / /
29 6 domain/lessons.ts

3.11 开源整合的真实状态（如实记录）

Human tlas `ashemag/human-atlas` MIT © 2026 ashemag
BodyParts3D CC BY 4.0 three ^0.159 React Three
Fiber three 0.186 / R3F 9 33 MB BodyParts3D ID Z-Anatomy
ID 45–70

- ✦ Human Atlas

natomy `thebuggeddev/anatomy` LICENSE / COPYING
package.json public/models/*.glb / / / / / / /
Z- natomy CC BY-S 4.0 +

Z- natomy / Vanatome GLB CC BY-SA 4.0

4.1 技术指标（本机真实执行）

---	---
tsc -b	**0**
oxlint	**0 error / 14 warning**
vitest run	**90 / 90** domain 22 + cellGeometry 19 + fullBody 29 + structureCard 8 + organBridge 12
npm run build tsc -b && vite build	
	337.46 kB / **107.59 kB gzip**
OrbitControls	928.08 kB / **246.42 kB gzip**
FullBodyStage	85.44 kB / **26.49 kB gzip**
CellStage	11.16 kB / **4.09 kB gzip**
index.css	32.78 kB / **6.84 kB gzip**

4.2 逐页与交互实测

16 /about canvas = 0 15 canvas = 1

GLB **984** **17** 14 + 2 + 2 **967**
42% **734** body-shell **233**

/ Esc
/ / inert = true
body-day/scripts/build-docx.py

4.3 未完成的验证（如实记录）

9. **Human tlas** + 3.11
10. **natomy `thebuggeddev/anatomy`** **/**
11. 390×844 /
12. **GPU** SwiftShader FPS
13. = × 0.17
14. Firefox / Safari Chromium
15. + GLB
16. **/** urinary 3
17. /
18. 2026-10-18

5.1 环境要求

Node.js 18 v24.15.0 WebGL

5.2 安装与运行

```bash
cd body-day
npm install # npm ci
npm run dev # → http://localhost:5173/
npm run build # dist/
npm run preview # → http://localhost:4173/
npm run test #
npm run lint #
```

5.3 部署

HashRouter

public/models/ **/** VITE_BODYDAY_ATLAS_URL

5.4 操作说明

---	---
/	
	0.07 [0,1]
/	
/	
	/ /
	/ /
→	/ / /
/ Esc	
Tab / Enter / Space	

6.1 已完成

+ / /

/ / /

organBridge / / +

17 → → ↔ 14 3 29

6 **90** **16**

6.2 主要技术难点与处理

19. atlas

position

20. R3F GLB extras +

+

21.

22. /about canvas = 0

6.3 不足与后续升级

GPU tissue +

/ urinary 3

Human Atlas MIT BodyParts3D CC BY 4.0 three /

R3F 33 MB BodyParts3D ID 45–70

+ Anatomy thebuggeddev/anatomy

Z-Anatomy + → β

V2 delivery/demo/bodyday-v2-walkthrough.mp4

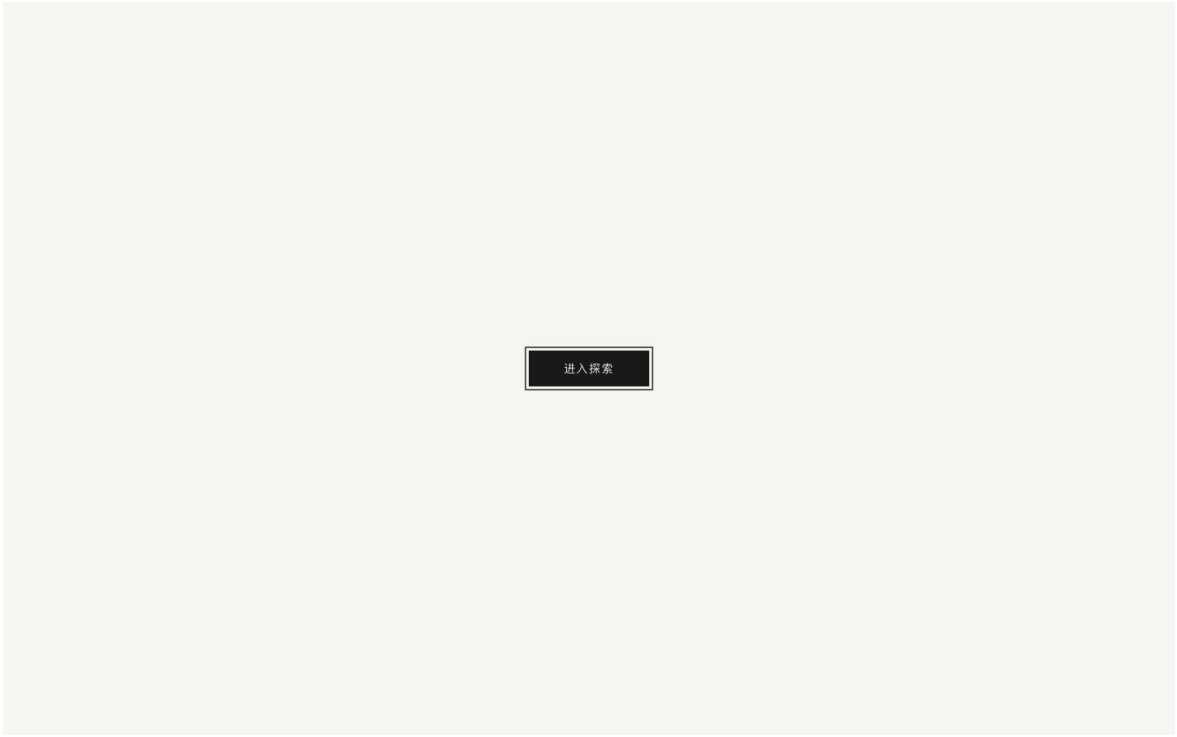
6.4 □□□□

BodyDay

SOURCES.md 19 2026-10-14

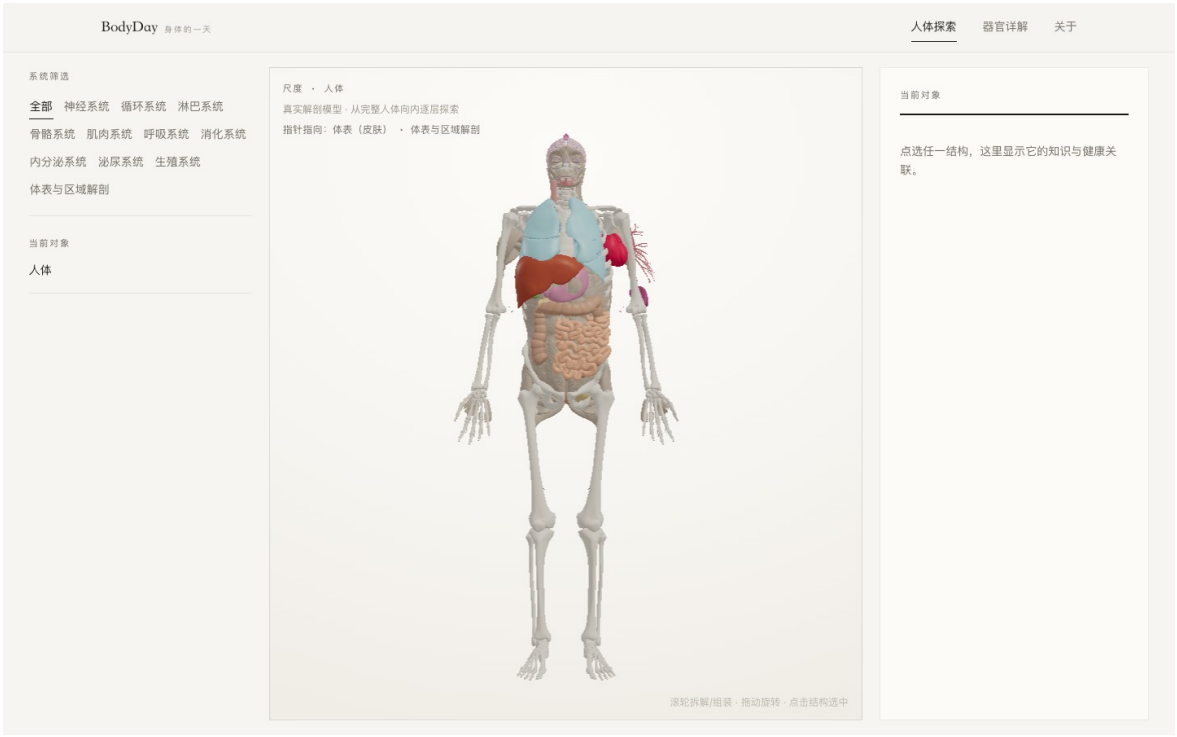
-

1440×900　　390×844　2x

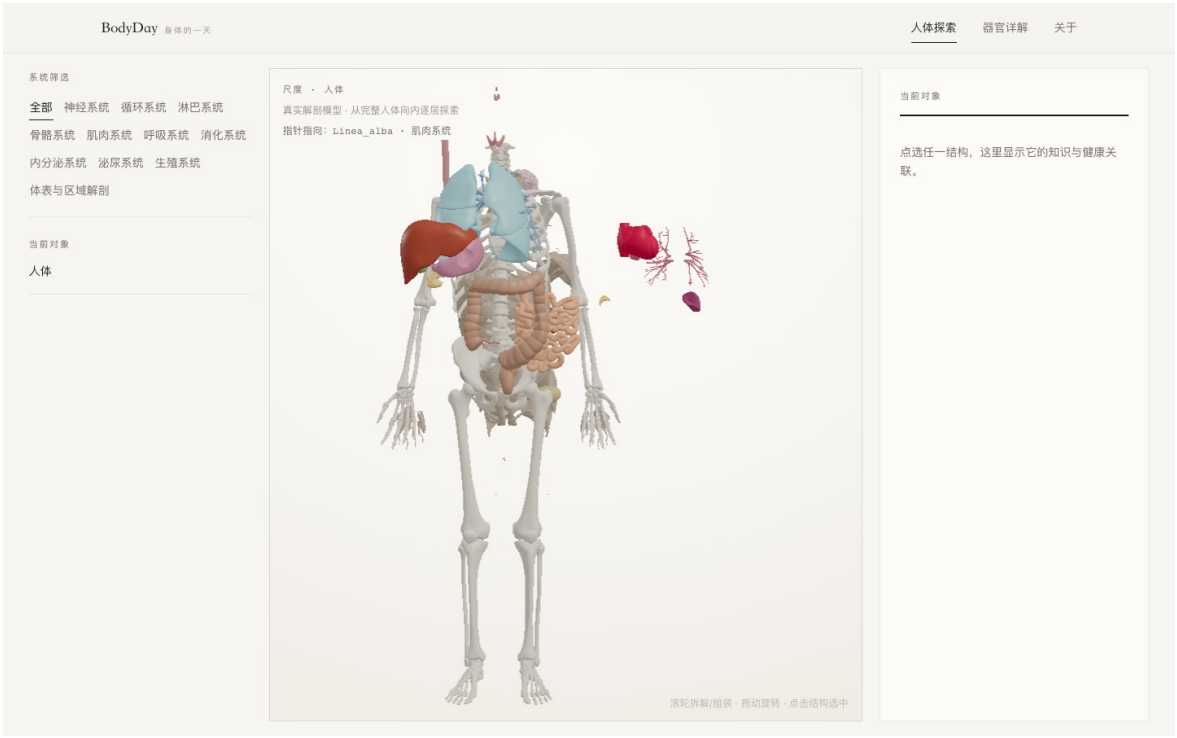


01-cover

02-body



03-explode-25 25%

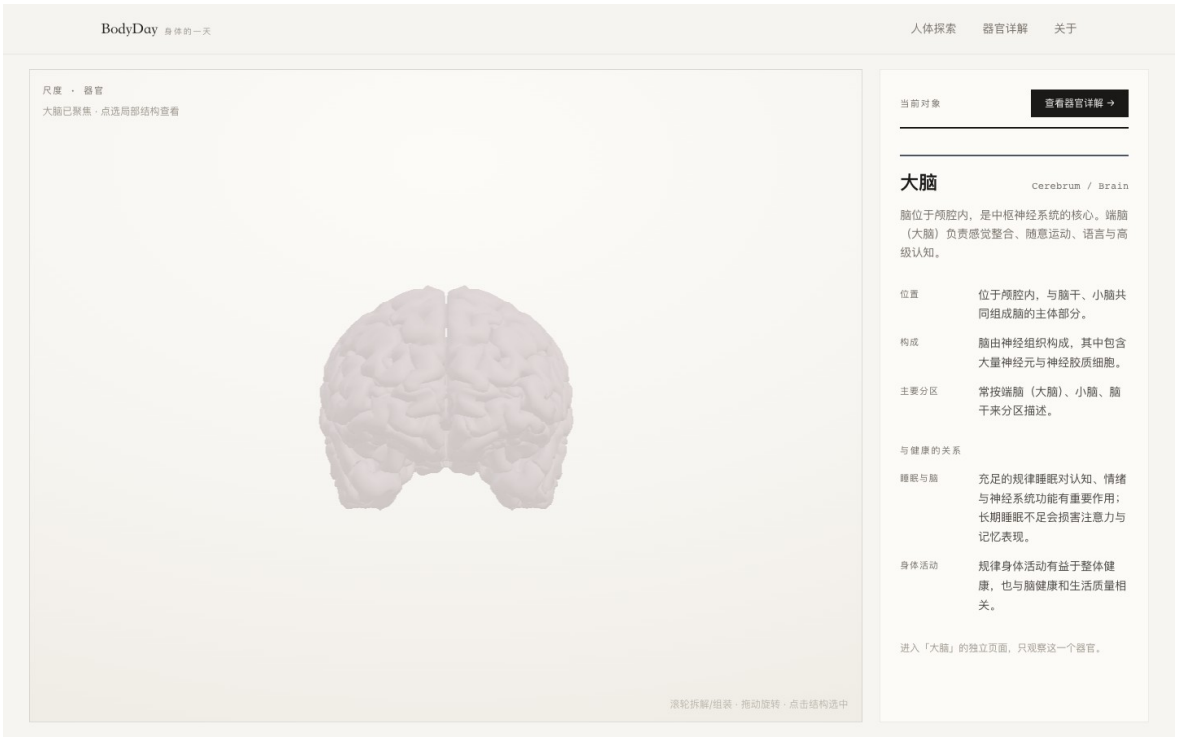


04-explode-100 100%

05-system-nervous

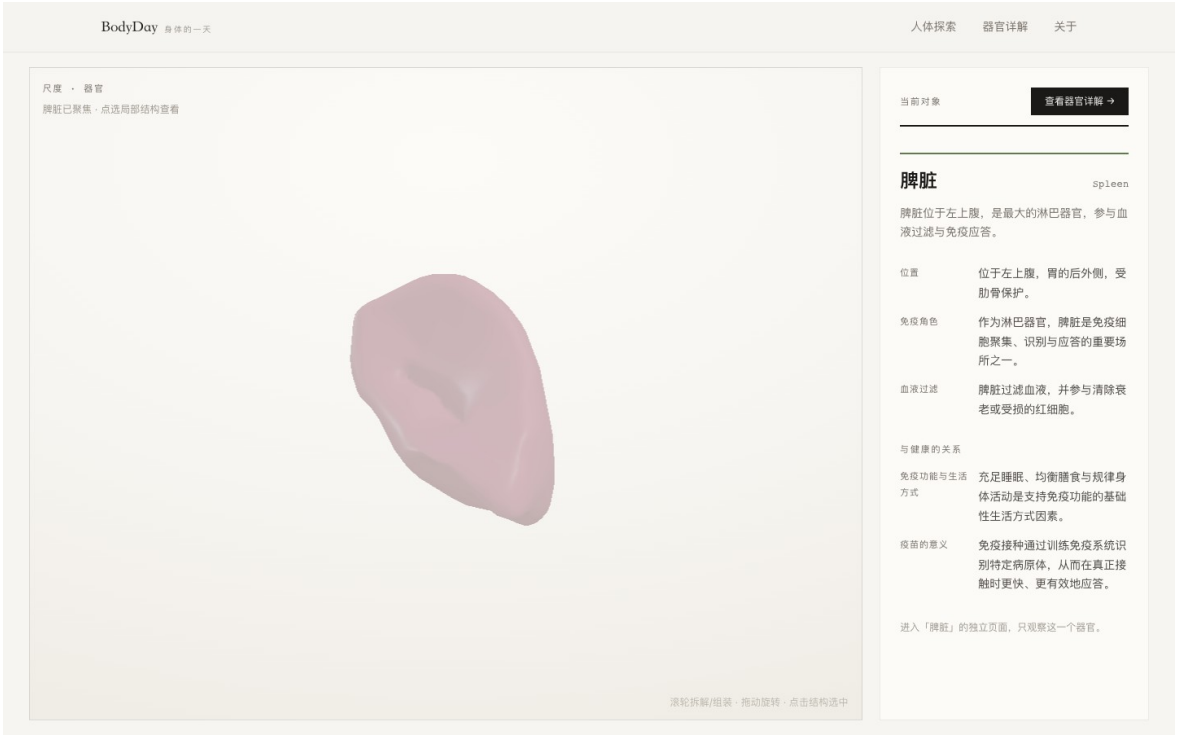
06-system-cardiovascular

07-system-lymphatic



08-organ-brain +

09-organ-heart +



10-organ-spleen +

11-cell-neuron

12-cell-leukocyte

BodyDay 身体的一天
人体探索
器官详解
关于
首页 / 关于
11 · 关于 · 数据与方法
BodyDay 是怎么做出来的
把「来源、边界、许可」写在最后一页，是因为科普作品的公信力
和它的画面一样重要。
方法
内容是怎么被组织起来的
结构化数据
实体 16 条、科学关系 14 条、来源 19 条， 全部为纯数据，可逐条核对与扩展。另附跨尺度科普脚本 3 套， 作为内容资产保留，供后续扩展使用。
真实模型优先
人体与器官层只使用 atlas 中**真实存在**的结构 ID（如 cerebrum、heart、spleen）； 聚焦前会先确认该 ID 存在于模型元数据中。
关系语义化
结构隶属、功能相关、循环输送、疾病语境分别登记，关系卡片永远写出自己的科学语义， 不使用自动兜底关系。
来源可追溯
每条知识、每道检验题都带来源编号；来源表列出出版者、标题、原始链接与访问日期。医学细节若无法核对，标注为待

尺度 · 人体

真实解剖模型 · 从完整人体向内逐层探索

14-mobile-390x844 390×844